

Load regulation. It defines the consistency in the output even if the load changes. If the load impedance changes, ideally the set parameters should not change significantly. Load regulation is mentioned in the data-sheets as $\pm$ (% set value + offset value).

Line regulation. It defines the consistency in the output even if the input supply or frequency to the power supply varies. It is represented in the same way as load regulation.

AC characteristics

Ideally, we expect pure DC at the output but there is always some AC component present together with DC component as no power supply is perfect. Some devices are highly sensitive to this AC component.

AC characteristics defining specs are noise and ripple, and transient response.

Noise and ripple. Ripple is a periodic component, whereas noise is completely random. These components are not indicated separately as they cannot be separated. Noise and ripple is expressed as a root-mean-square as well as peak-to-peak value.

Transient response. Transient response indicates the response time of power supply to changes. If you change the voltage, transient response indicates how quickly the power supply settles to the set value.

Packaging

Packaging gives details like overall dimensions and ports location. Most power supplies are designed for standard 48.3cm (19-inch) rack dimensions.

Features to look for

Features that are highly useful in lab experiments include:

Remote sense. It ensures that the exact expected voltage is delivered to the load. Some amount of output voltage is always dropped in the cables and the voltage to the load is always less than the meter indication. This feature is implemented in power supplies by using another pair of ports

to read the voltage at the load and auto-correct the power supply output accordingly.

Coarse and fine adjustment. Some power supplies do not have fine adjustment knob, which is a must have for all the testing tasks. This knob helps to set fine output values, say, 12.3V. It would be very difficult to set such a value just with the main coarse adjustment knob.

Voltage span. With voltage span feature you can limit the full span of the coarse knob to a voltage range. There are some applications that are very sensitive to the operating voltage range. For such applications, it is always best to limit the full span of the knob within the operating range.

Output-enable button with default 'off.' This is a small but very critical feature for testing electronics applications. Output-enable button enables voltage from the power supply to the load. With this feature, you can comfortably set output voltage and enable the output only when you are done. Otherwise, the change in setting will directly impact the load. When you turn on the power supply, the output should be 'off' by default.

Over-current drawn indication. Power supplies with this feature indicate whether the load draws more current than the set value.

Lock. This feature disables the operation of all the buttons. You can set the desired parameters and enable lock so that the parameters do not get changed by mistake.

Programmable. A programmable power supply can be interfaced to a computer with interfaces like GPIB (general-purpose interface bus) and RS232. Sometimes these are also interfaced to the network using Ethernet or USB interface. Using programming languages specified by the manufacturer, you can send commands to the power supply and control it. This feature is highly useful while developing automatic test equipment. ●

The author is technical editor at EPY

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